25 May 1955
SUPERSEDING
NAVORD OS 5979
6 April 1952

MILITARY SPECIFICATION

DETONATOR MARK 50 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, assembly, loading, and preparation for delivery of the Detonator Mark 50 Mod 0 and the methods of inspection and tests upon which acceptance will be based. The Detonator Mark 50 Mod 0 is flash initiated.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-T-339 -- Tetryl (Trinitrophenylmethylnitramine).

STANDARDS

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 166701 -- Detonator Mk 50 Mod 0.
LD 275959 -- Detonator Container Mk 6 Mod 0 (Inner).
LD 275960 -- Detonator Container Mk 8 Mod 0 (Outer).
And all documents listed on the above LD's.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) obtainable from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above should cite "49 CFR 71-78 (Rev. 1950).")

3. REQUIREMENTS

3.1 General.

3.1.1 Detonators covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings LD No. 166701 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Lead azide. -- The moisture content of this lead azide at the time of loading shall not exceed 0.30 percent when tested as specified in 4.5.1.

3.3 Tetryl. -- The volatiles and moisture content of the tetryl shall not exceed 0.1 percent when tested as specified in 4.5.2.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of Military Standard MIL-STD-105.

4.1.1 Lot. -- As applied to the Government inspection of units of product, the term "Lot" shall mean "Inspection Lot, i.e., a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

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4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of one thousand (1000) Detonators Mk 50 Mod 0 shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These detonators shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot. -- For procurement purposes each lot of loaded detonators shall consist of 8,000 units plus those required for all tests.

4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government inspection shall be in accordance with the provisions of General Procedures for Government inspection and Contractor's inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.5 Tests.

4.5.1 Moisture content test for lead azide. -- Conduct at least one test for each lot of loaded detonators and such additional tests as the inspector considers warranted by variations in the process and materials. Collect an approximate two-gram sample in a tared, stoppered container at the site of and immediately prior to consolidating the charge in the detonator. Weigh the container; open and expose the sample for drying at 50 ± 2 degrees c and at least 28 inches (Hg) vacuum until constant weight is reached. Cool the container in a desiccator; stopper and weight. All weighing shall be accurate to ± 0.5 milligram. The presence of moisture in excess of 0.30 percent shall cause suspension of the charge loading until permanent corrective measures satisfactory to the inspector have been completed. Detonators loaded with the defective charges shall be rejected.

4.5.2 Volatiles and moisture content test for tetryl. -- Conduct at least one test for each lot of loaded detonators and such additional tests as the inspector considers warranted by variations in the process and materials. Test procedure shall be in accordance with that given in Military Specification JAN-T-339. The presence of volatiles and moisture in excess of 0.10 percent shall cause suspension of the charge loading until permanent corrective measures satisfactory to the inspector have been completed. Detonators loaded with the defective charges shall be rejected.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. -- One thousand (1000) detonators shall be packaged in accordance with Bureau of Ordnance List of Drawings LD No. 275959.

5.2 Packing. -- Four thousand (4000) detonators, packaged as specified in 5.1, shall be packed in accordance with Bureau of Ordnance List of Drawings LD No. 275960.

5.3 Marking. -- In addition to any special marking required by the contract or order all containers shall be marked in accordance with Military Standard MIL-STD-129 and Code of Federal Regulations 49 CFR 71-78.

5.4 Data cards. -- Data cards, five inches by eight inches, printed as shown below, shall be filled out for each lot of detonators. Two of these cards shall accompany the shipping ticket; three shall be forwarded to the Procuring Agency, and five shall be furnished the inspector.

AMMUNITION DATA CARD

Mark No._____ Lot No._____

Dwg. No._____ Spec. No. _____

Loaded By_____ Date (Mo. and Yr.)_____

Contract No._____ Req. No._____

No. in Box_____ No. in Lot_____

MATERIAL	MANUFACTURER	LOT NO.	DATE (MONTH, YEAR)
CUP			
CLOSING DISC			
SEALING DISC			
LEAD AZIDE			
TETRYL			

Remarks: (These are to include any changes in materials or manufacturing processes prescribed in this specification which the inspector feels might affect the uniformity of the lot).

6. NOTES. -- There are none applicable.

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Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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DOCUMENT IDENTIFIER AND TITLE

MIL-D-18681(NOrd), Detonator Mark 50 Mod 0

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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